

An Enterprise Imaging Strategy for Ophthalmology

EAS 5850: Imaging Informatics — Homework 4

Introduction

Enterprise imaging (EI) is the set of strategies, policies, and technology used to capture, store, manage, view, and share medical images across an entire health system. The goal is one source of truth. Every image should be tied to the right patient, stored in a common archive, and viewable from the electronic health record (EHR). I have only worked in radiology, so my first step is to learn how ophthalmology differs before I copy a radiology design onto it.

Radiology is a mature, standardized field. Almost every study is DICOM. Every study begins with an order, so it has an accession number and a clear link to a patient and a reason. This is an *orders-based* workflow. Ophthalmology is different. Images are captured during a clinic visit, often without a formal order, so it is an *encounters-based* workflow. Ophthalmology also uses many small devices from many vendors. Many of these devices do not fully support DICOM and export proprietary or non-DICOM files. The result is a high volume of varied data that is harder to standardize than a radiology PACS.

Data Considerations

Type of data. Ophthalmology produces many modalities: color fundus photography, optical coherence tomography (OCT) and OCT angiography, visual field (perimetry) reports, slit-lamp and external photographs, fluorescein and ICG angiography, corneal topography, and biometry. Some of this is true DICOM (Ophthalmic Photography and OPT objects), but much of it is non-DICOM: JPEG and TIFF images, PDF reports, and vendor-proprietary formats. A core task is to convert or wrap non-DICOM data so it can be managed in one place.

Storage of data. I recommend a Vendor Neutral Archive (VNA) shared across all six hospitals instead of separate departmental PACS silos. The VNA must hold both DICOM and non-DICOM objects, using DICOM wrapping or an XDS-style document store where needed. Because OCT volumes are large and image counts are high, storage should be tiered (fast storage for recent studies, cheaper storage for old ones) with a clear retention policy.

Viewers. Clinicians need a zero-footprint enterprise viewer that opens inside the EHR, shows color correctly, and supports basic ophthalmic tools. Some diagnostic tasks still need the native device software (for example, OCT trend analysis), so the strategy must allow specialty viewers alongside the universal one.

Clinician and patient access. Viewing should launch from the EHR in patient and encounter context with single sign-on, so no separate login is needed. Patients should reach their images and reports through the patient portal.

Workflow Considerations

The hardest workflow problem is the lack of orders. Without an order, an image can arrive with no reliable patient link. I would push devices to use a modality worklist where possible, and for devices that cannot, use barcode patient identification at the point of capture. An enterprise master patient index (EMPI) is needed so the same patient is matched correctly across all six hospitals. Images must be reconciled to the correct patient and encounter before they reach the archive.

Cost and Return on Investment

A shared VNA and viewer cost money up front, plus integration effort. The return comes from removing duplicate PACS systems, lowering long-term storage cost, reducing vendor lock-in, and cutting repeat

imaging because prior studies are easy to find. Better data capture also improves billing and supports research and future AI tools.

Impact on Patient Experience

Patients benefit because their images follow them across all six hospitals. They face fewer repeat scans, faster results, and portal access to their own data. Care is more continuous because any provider in the system can see prior imaging.

Impact on Clinician Wellness

A single viewer in the EHR means fewer logins and less hunting through separate systems. A standard workflow reduces clicks and frustration, which helps lower burnout for both ophthalmologists and support staff.

Governance Considerations

I would form an EI governance committee with ophthalmology, radiology, IT, security, and compliance representatives. The committee sets metadata and naming standards, decides retention rules, defines who owns the data, and prioritizes work fairly across all six hospitals so no single site is favored.

Implementation Considerations

I recommend a phased rollout. First, inventory every imaging device and check its DICOM capability. Second, select the VNA and enterprise viewer and integrate them with the EHR using HL7 and FHIR. Third, pilot at one hospital, fix the device and reconciliation issues there, then expand site by site. This limits risk and lets staff learn before the full rollout.

Cybersecurity Considerations

All data is PHI, so it must be encrypted in transit and at rest, with role-based access control and full audit logging. Many imaging devices run old, unpatched software, so they should sit on a segmented network. Across six sites I would standardize patching, secure remote access, and a tested backup and disaster recovery plan, all aligned with HIPAA requirements.